

Big data & HR

A practitioner's perspective

“I was looking for something like baseball,
where there's a lot of data and the
competition was pretty low.”

Nate Silver, Guardian interview

Data Scientist

Job ID	281649	Location	US-WA-Seattle
Posted Date	3/2/2015	Company	Amazon Corporate LLC
Position Category	Business Intelligence		

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Job Description

“If you’ve never considered applying your talent in data sciences to the field of HR, you should: HR Analytics is a rapidly growing area of research and application where you can really make your mark.”

- Interact with executive project sponsors across multiple businesses to understand the business outcomes important for success; operationalize those outcomes in ways that can be measured and ultimately predicted.
- Drive the collection of new data and the refinement of existing data sources.
- Interpret data and communicate complex findings to leaders in HR and across business.
- Write research papers for internal audiences.

Basic Qualifications

- 8+ years of industry experience
- Master’s degree in Industrial - Organizational Psychology, Behavioral Economics, Statistics, or related field
- Strong statistical skills (HLM, regression, ANOVA, IRT methods, Structural Equation Modeling, etc)
- Strong technical skills and proficient in programs such as SPSS, R, Excel, Access
- You document your approach for reference and reproducibility in the future
- Self-starter with the ability to work independently
- Great organizational skills and attention to detail; you are able to prioritize multiple tasks simultaneously without sacrificing the ability to dive deep
- Ability to communicate complex quantitative analysis in a clear, precise, and actionable manner
- Impeccable customer service focus with a demonstrated desire to exceed expectations; a team player with a solid work ethic

Preferred Qualifications

- PhD strongly preferred in Industrial - Organizational Psychology, Statistics, Behavioral Economics, Statistics, or related field
- Experience with RedShift and/or Hadoop; preferably with design and deployment
- Previous work experience in applying statistics to HR and business outcomes

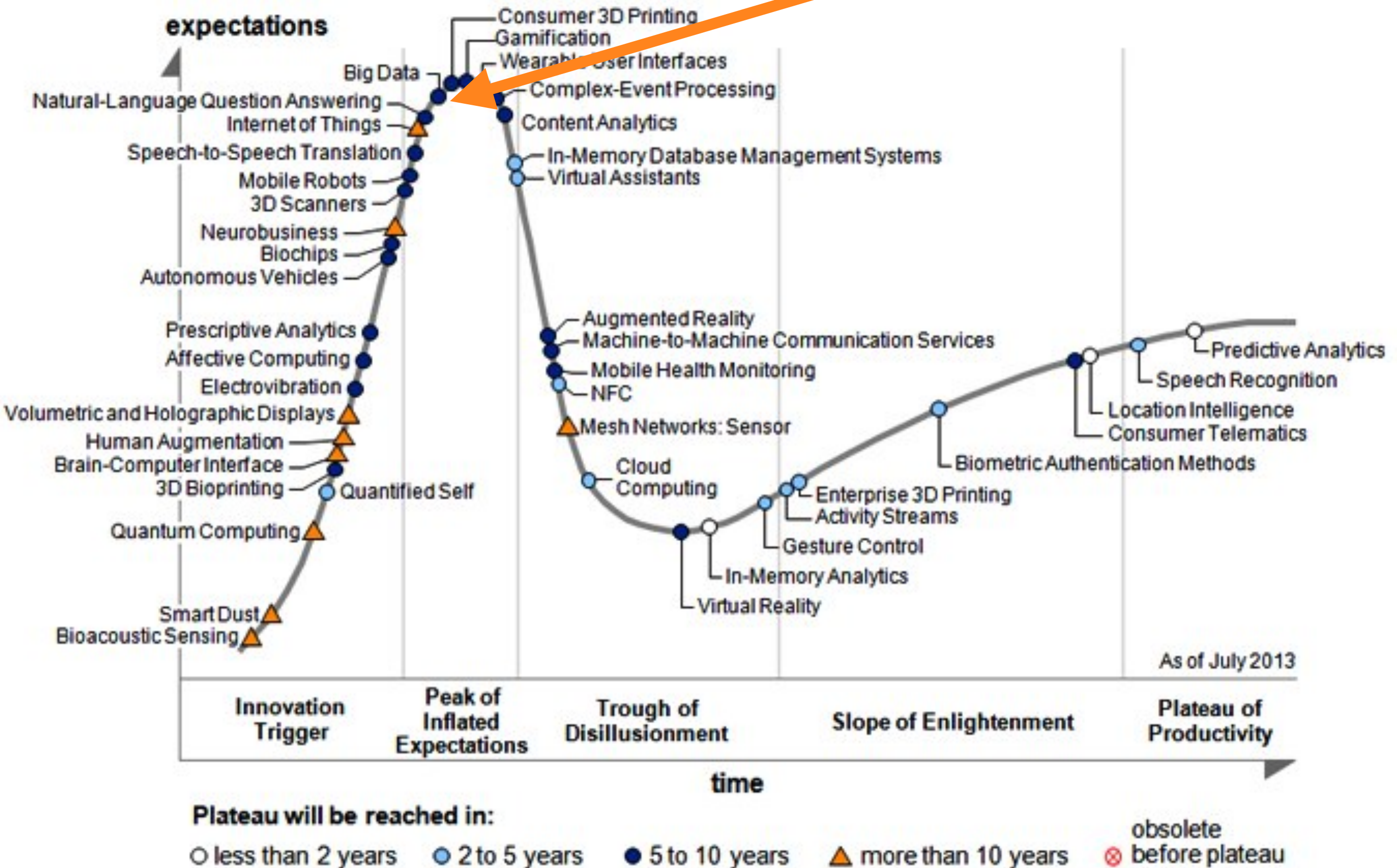
WHAT IS BIG DATA?

Big data is a bit like teenage sex:
Everyone talks about it, nobody really
knows how to do it, everyone thinks
everyone else is doing it, so everyone claims
that are doing it.

Dan Ariely

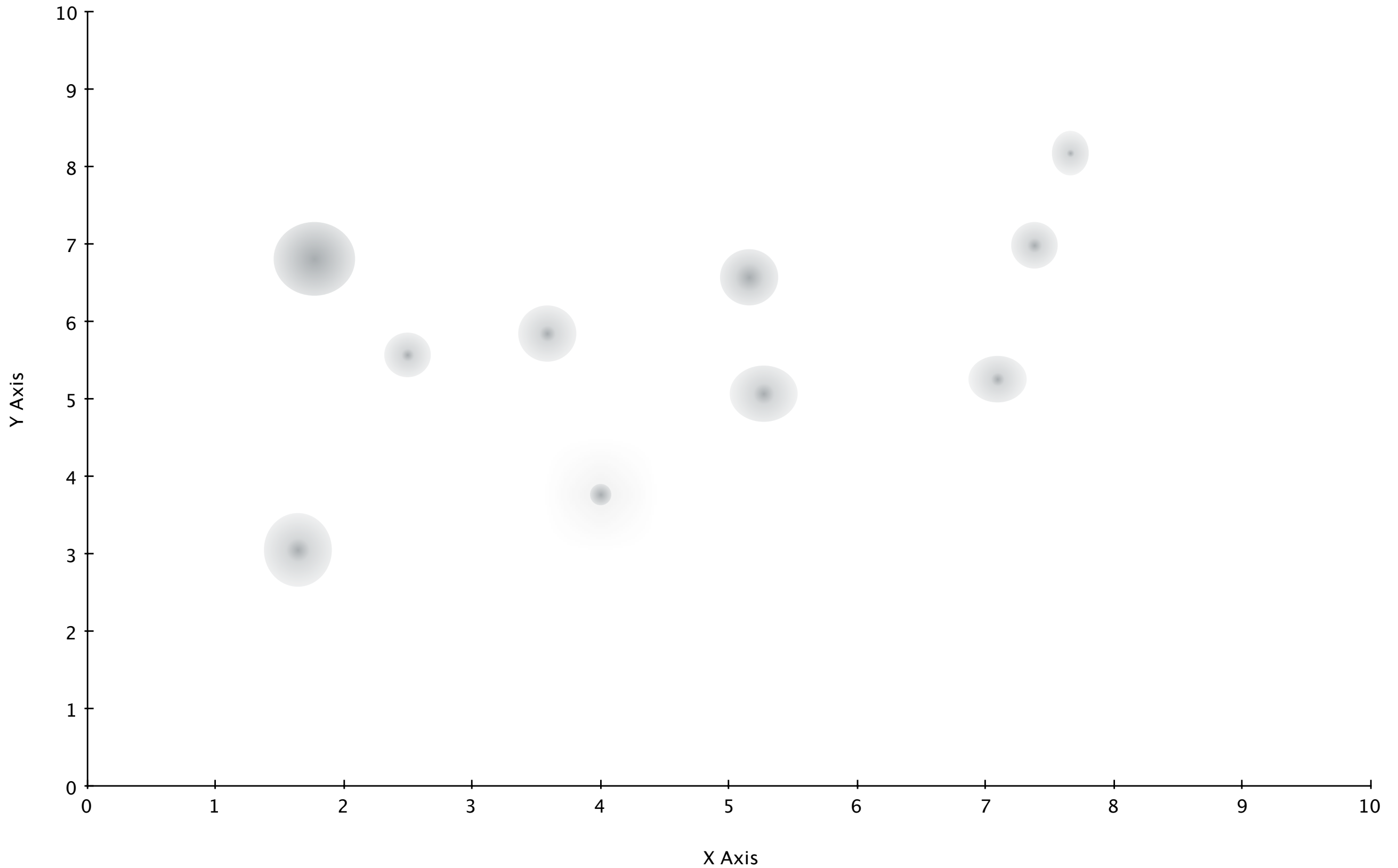
Gartner 2013 Hype Cycle

Big Data



BigData <- Volume +
Velocity +
Variety +
Veracity

Veracity in data



Open text offers lots of promise to HR

Everyone is leaving and getting much higher pay, but we are told we are paid at market rate. I think your data is wrong

I haven't had a career discussion in years

Management only communicates that it wants us to change

I don't see any change from last year

The new intranet is a huge improvement

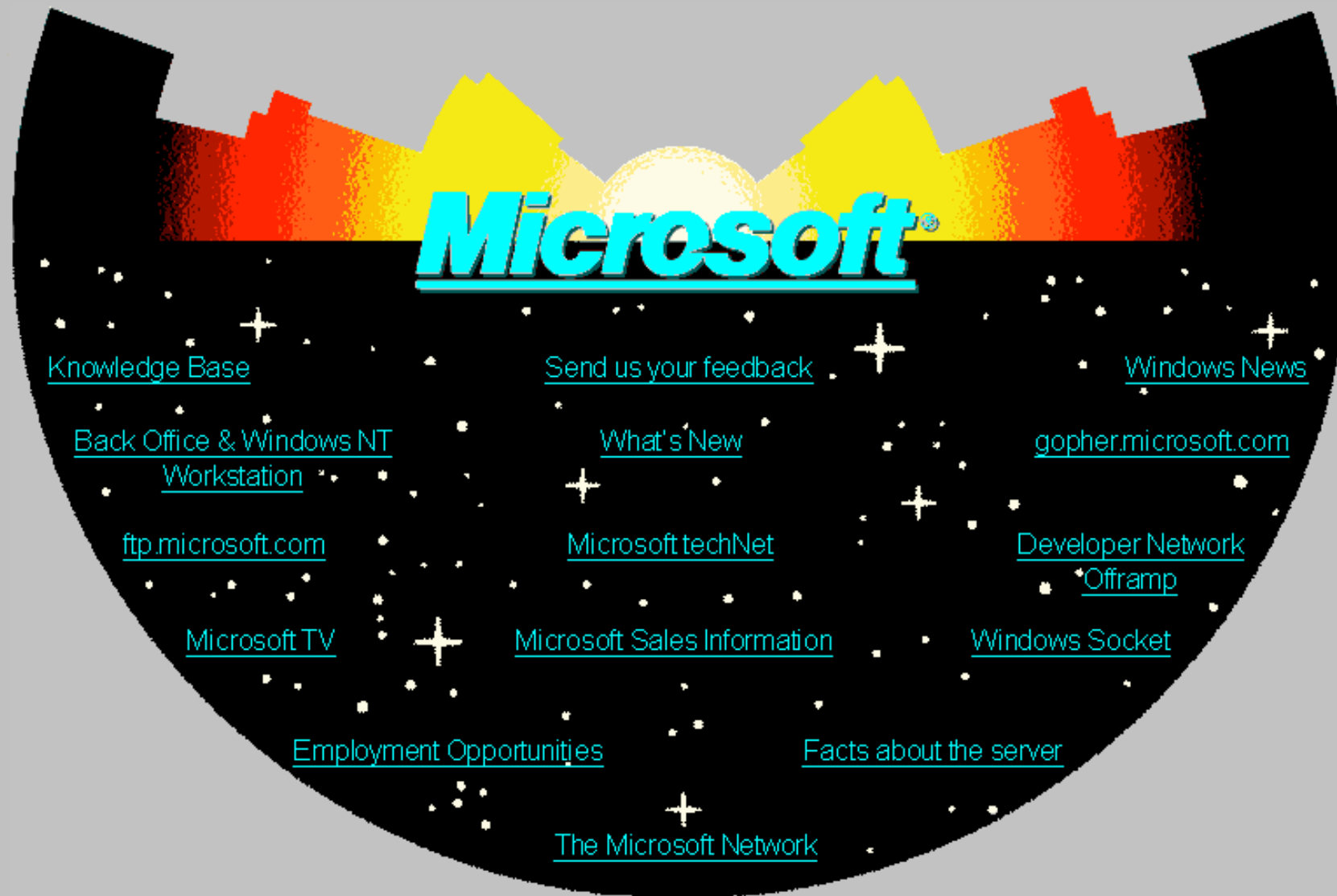
Our coffee machine has been broken for ages. We have to walk 5 minutes to get a drink.

Technology is changing rapidly

Welcome to Microsoft's World Wide Web Server!

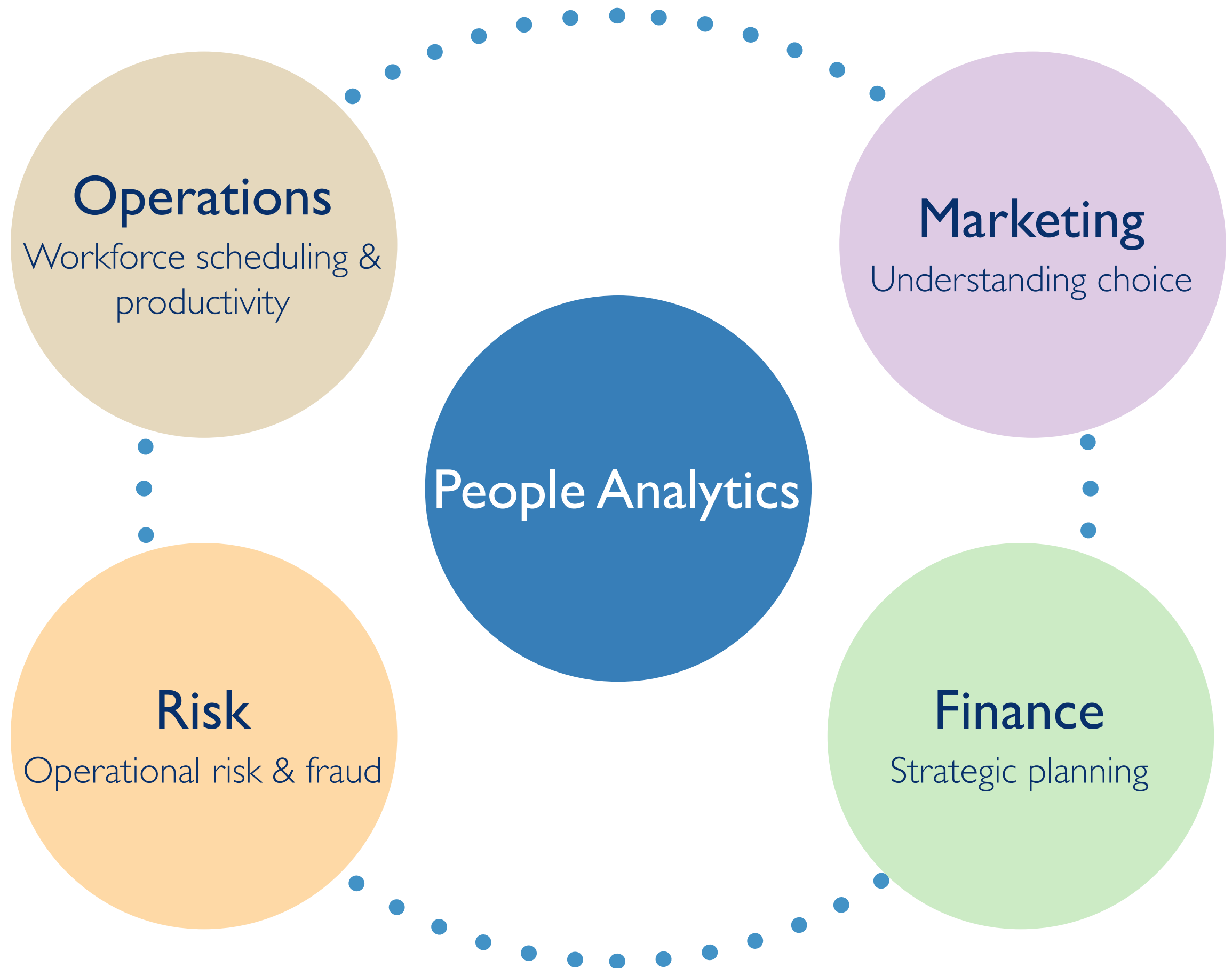
Where do you want to go today?

If your browser doesn't support images, we have a [text](#) menu as well.



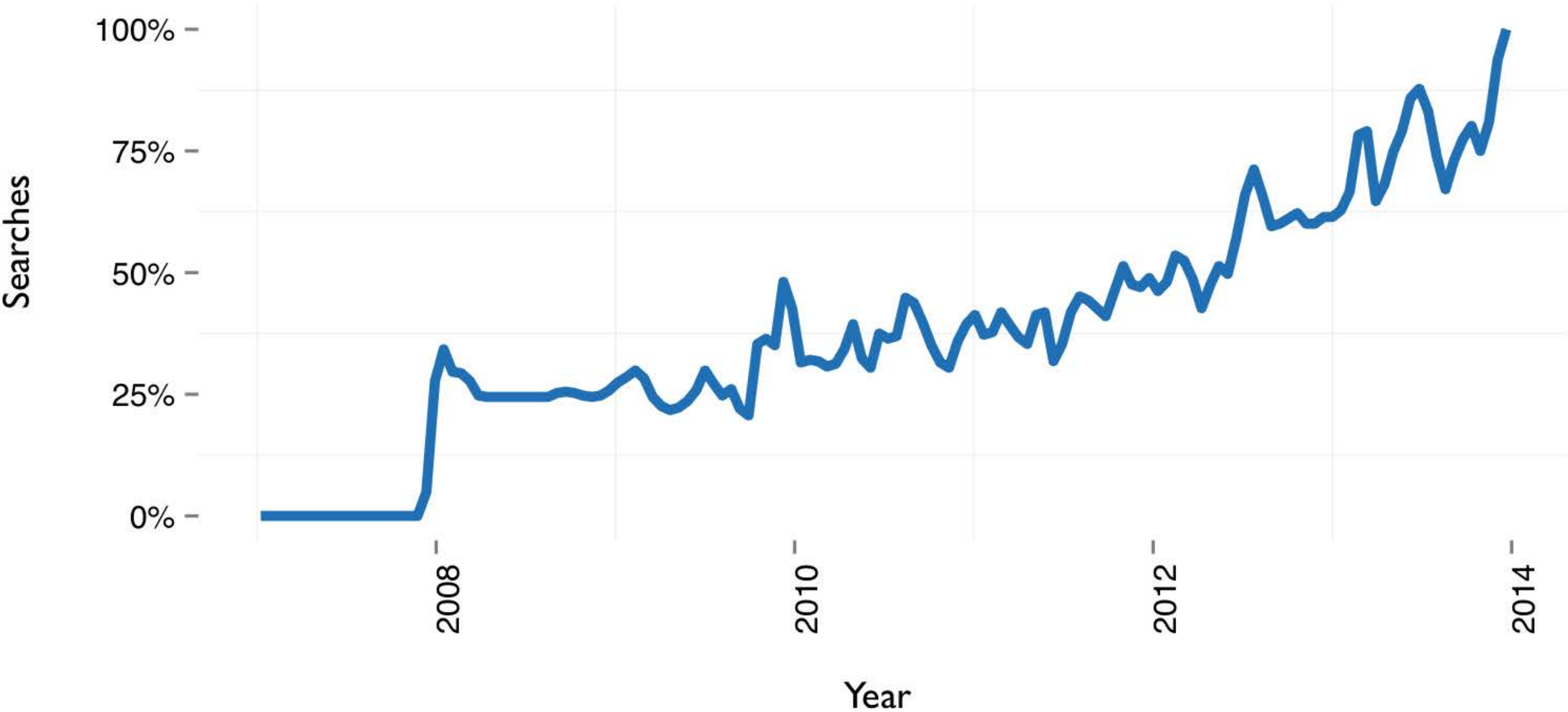
WWW.MICROSOFT.COM is running Microsoft's [Windows NT Server 3.5](#) and [EMWAC's](#) HTTPS

BIG DATA IN HR = PEOPLE ANALYTICS



Growth in the interest of HR Analytics

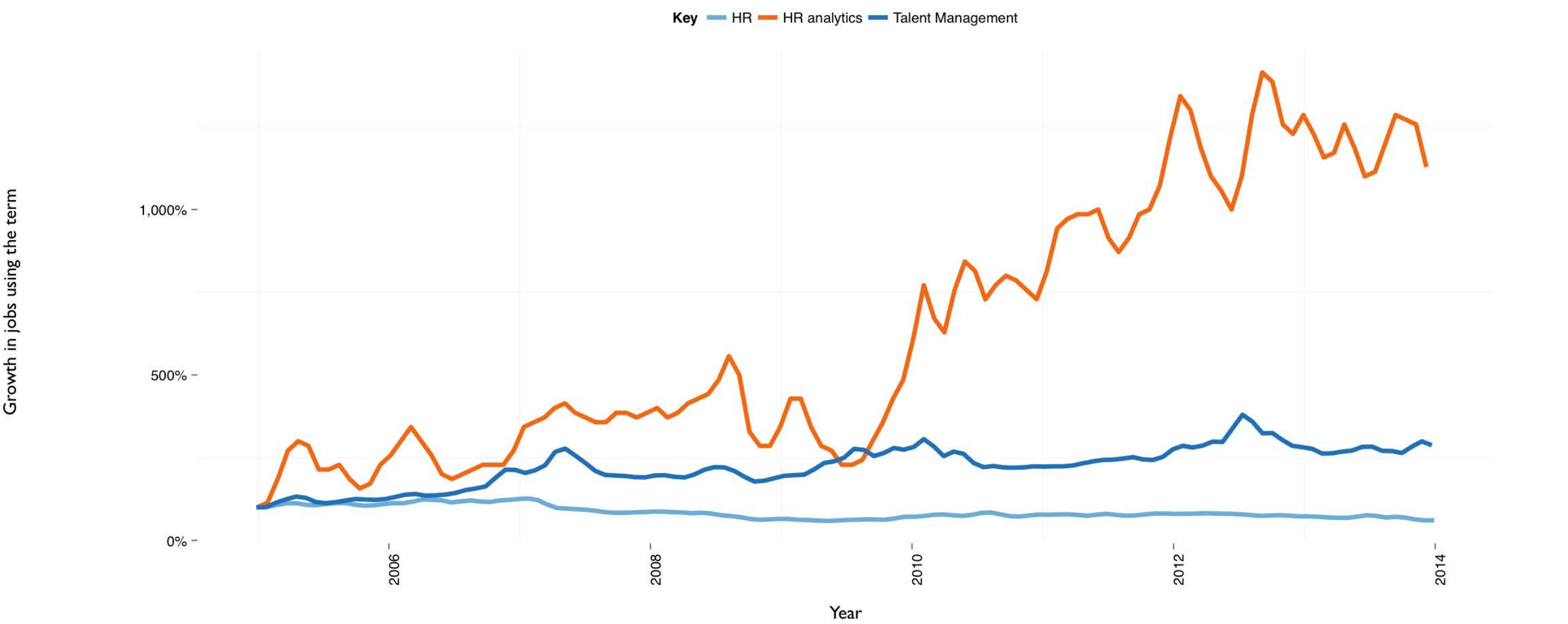
Web searches on Google as a percentage of peak period



Data: Google Trends, captured 31/10/14.

Growth in the jobs advertised for HR analytics

Occurrence of the following terms in job advertisements. Values were originally percentage of total jobs and were changed to index values based Jan 2005



Data: Indeed.com job trends, captured 31/10/14. Data visualisation by OrganizationView

The impact it can make

Leading Analytic Organizations have 17% better talent outcomes than other organizations

Corporate Executive Board

High performing organizations have higher levels of sophistication in their use of analytics and are more likely to have adopted analytics across all aspects of human capital management

EY, CFO - CHRO report

Quantified organizations have 79% higher Return on Equity than non-quantified Organizations

Sierra-Cedar 2014 Survey

Only 15% of respondents use margin / profitability as their most important measure of HR effectiveness

Accenture research

5 groups of vendors

The Technologists

The Data Aggregaters

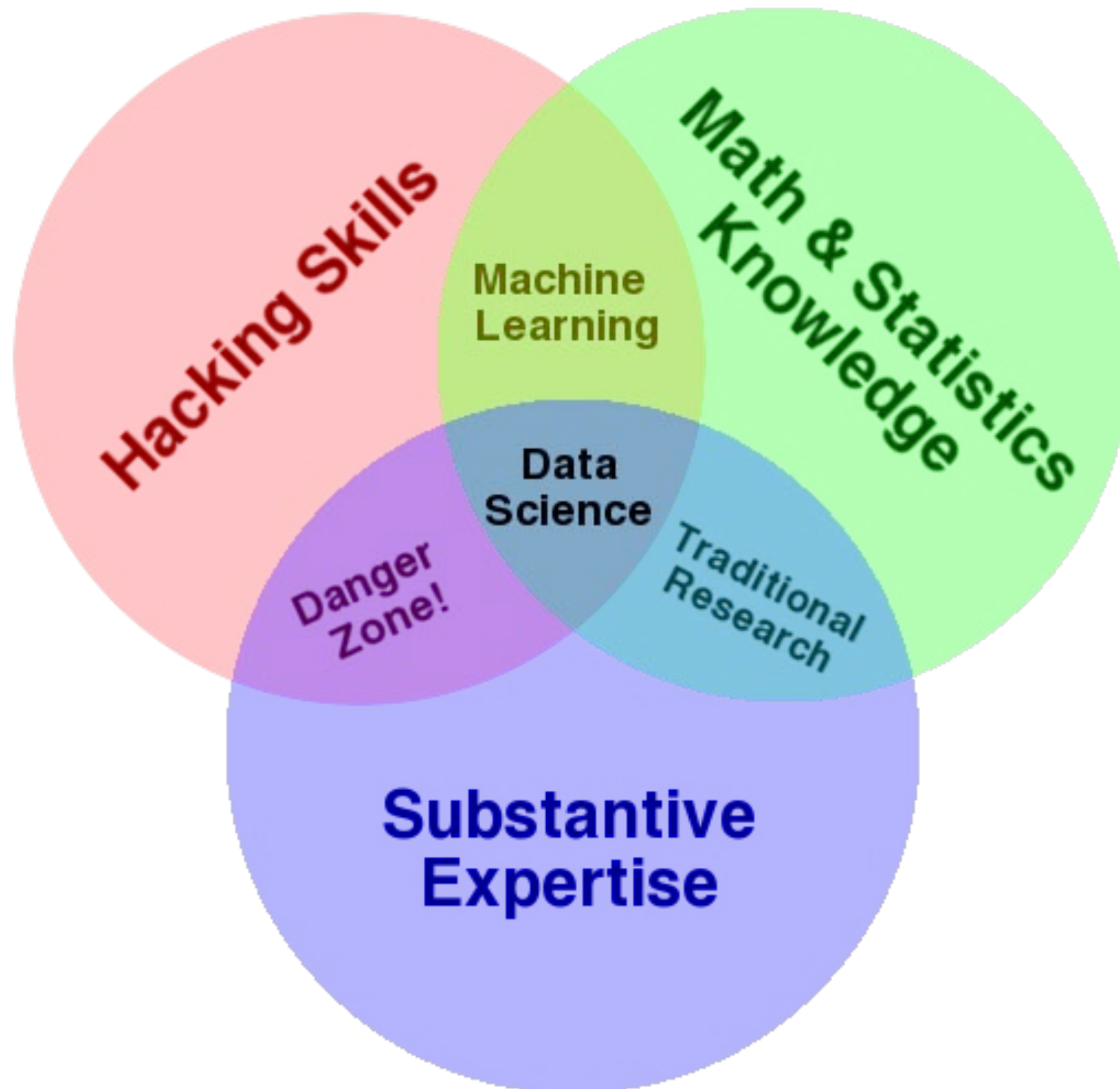
The Business / HR Consultancies

The Start-ups

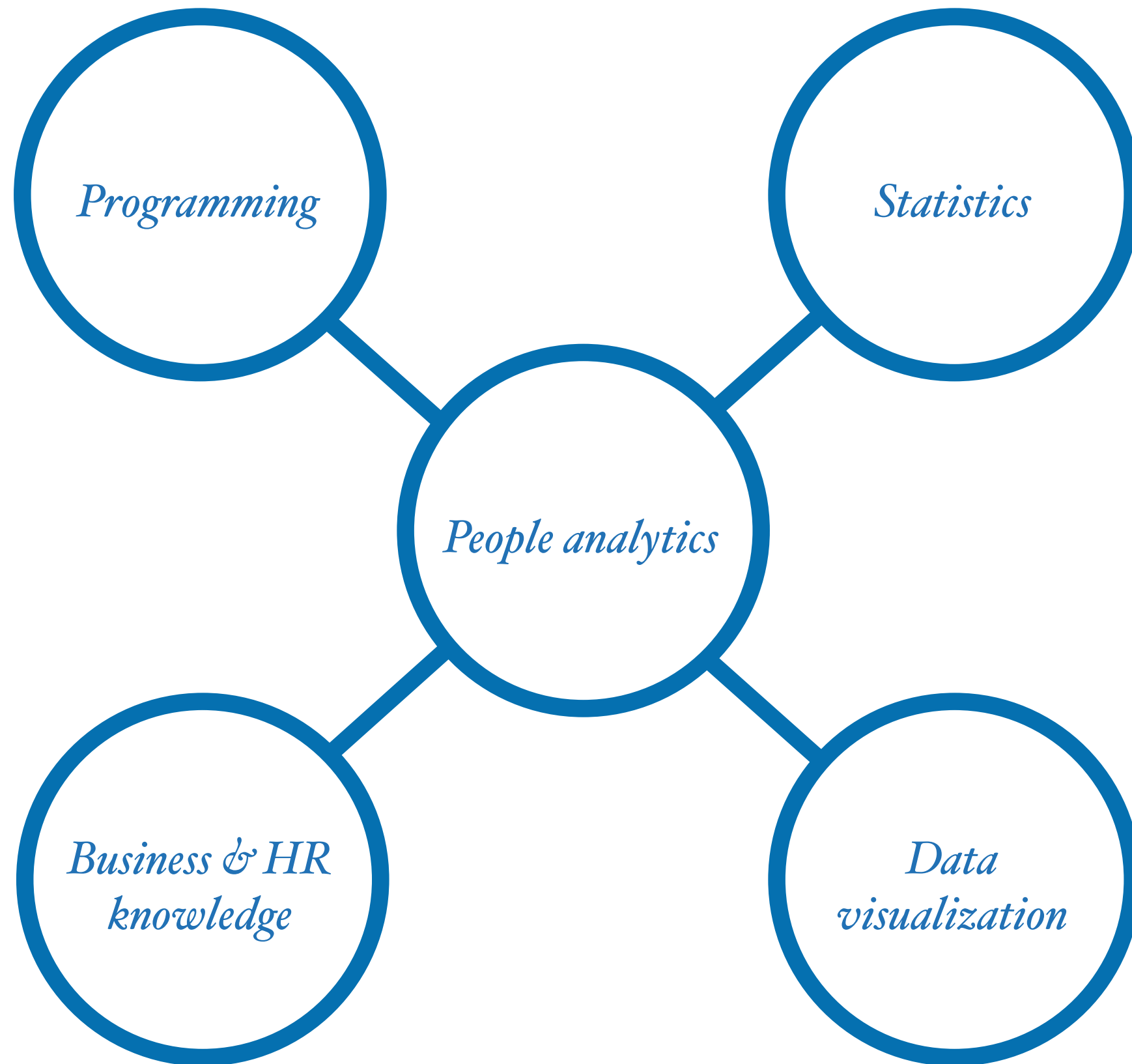
Analytics as a Service Providers

SKILLS

Drew Conway's Data Science Venn Diagram



What people analysts need





people analytics



About Google

Careers

Search all jobs

People Analyst, Tech HR

All Jobs

My Applications

Starred



☆ People Analyst, Tech HR

Mountain View, CA, USA

People Operations • Full-time

Know someone

APPLY NOW

At Google, data drives everything. As a People Analyst, you'll solve a strong problem so that everyone can get the top of it all, you're solving it with meaningful insights.

As a self-starter, you'll lead the next team leader the next day. With a background, an excellent aptitude for) thinking, technical skills, business insight that will allow you to perform.

Responsibilities

• Become an expert

Preferred qualifications

- BA/BS degree in Engineering, Mathematics, Statistics, Economics, or related technical field.
- Extensive experience with data visualization tools.
- Familiarity in one or more of the following areas: Coding (e.g., C/C++, Python, Java), Databases and querying (e.g., SQL, MySQL, MapReduce, Hadoop), Front end development (e.g., HTML, JavaScript, CSS), Statistical analysis (e.g., R, Stata, SPSS, SAS).
- Data-driven and highly analytical, and able to translate data and trends into powerful strategies and compelling messaging.



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Quantitative Analyst - People Analytics

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Please limit to 3
applications.

- 3+ years experience applying statistical methods to people research problems in a work environment, such as , Dimensionality Reduction Techniques, Categorical Analysis, Generalized Linear Models, Psychometrics, and Research Design, etc.
- Demonstrated competence in managing and analyzing structured and unstructured data using tools such as R, Python, or SQL (menu-driven tools are cool but not powerful enough for all you want to do)
- 1+ year of experience with data visualization using tools such as R, Tableau, Gephi, d3, etc.

on a broad spectrum of people issues: to optimize employee happiness, performance, growth and retention, and to attract outstanding talent to Facebook. You'll help redefine our team

Video Surveillance & Access
Control
(Menlo Park)

TECHNOLOGY

Technology options

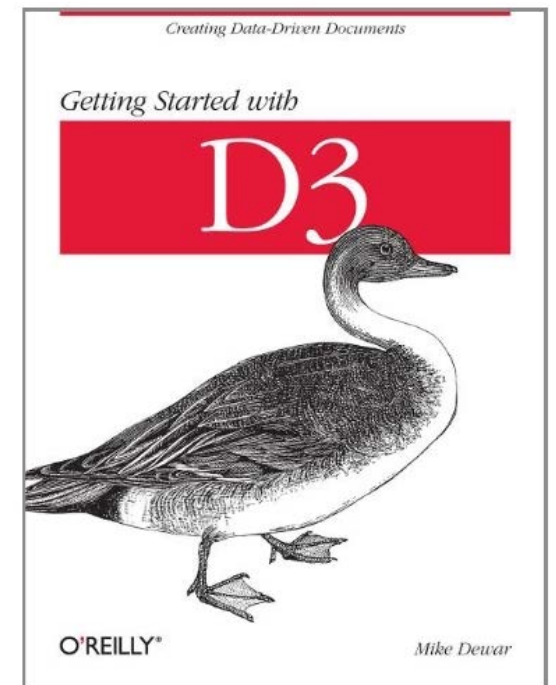
Embedded HR: analytics as part of HR technology

Standalone HR Analytics Tool: e.g. Visier

Enterprise BI: reporting tools such as Business Objects or Tableau or analytics tools such as SAS or R

Combinational Solutions: using a mixture depending on task.

Advanced teams are using mostly open source



GETTING STARTED

Always start here



Analytic Value Chain



Decisions based on
experience & gut
feel

Information is
structured but raw

Ratios, counts:

Can show trends
but audience gets
numb over time

Showing only when
needed reduces
fatigue

Correlations &
Relationships

Often misses the
'Why?'

Leads to action

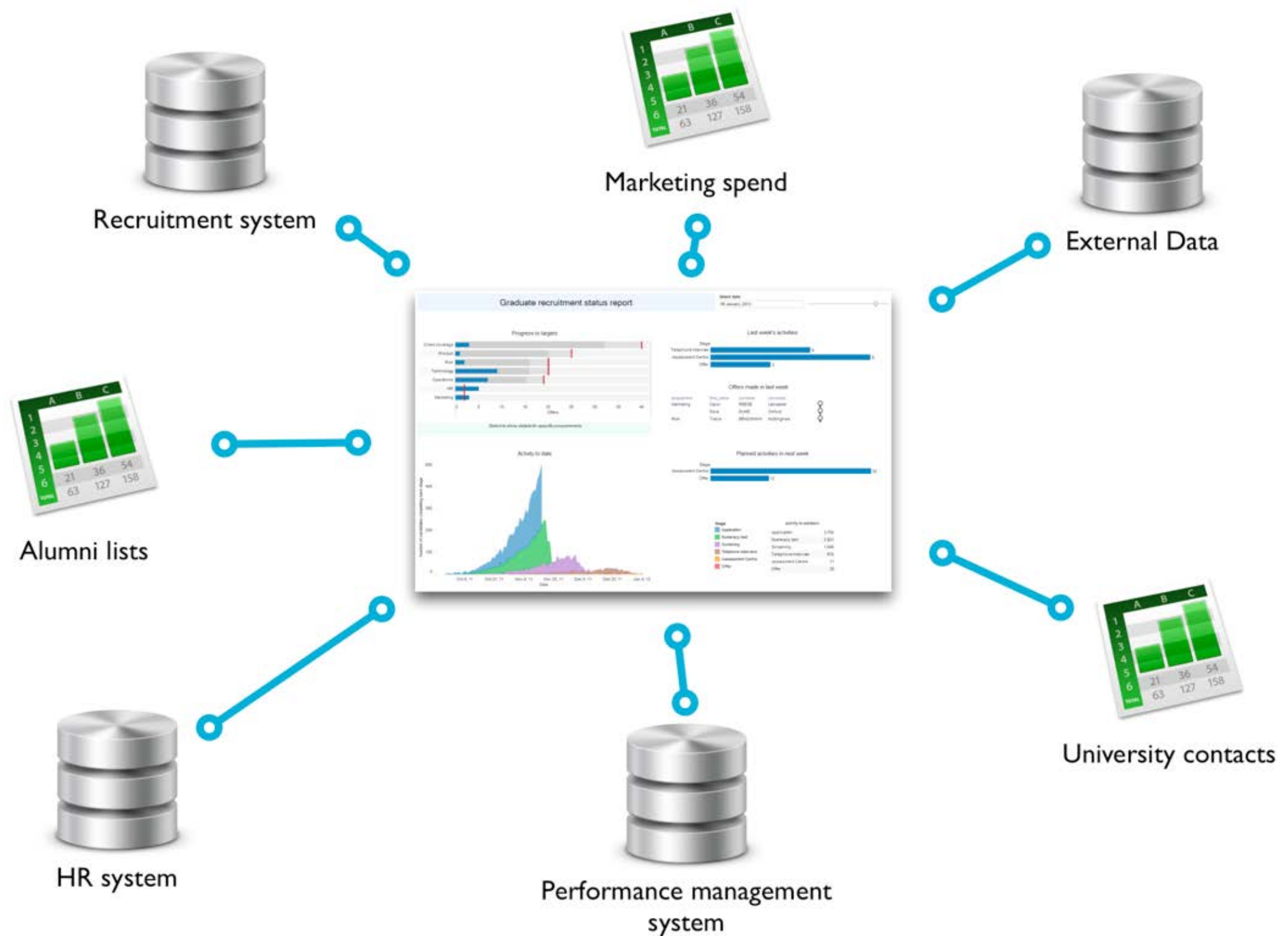
Needs to
incorporate a
narrative

Changes in
processes and
policies

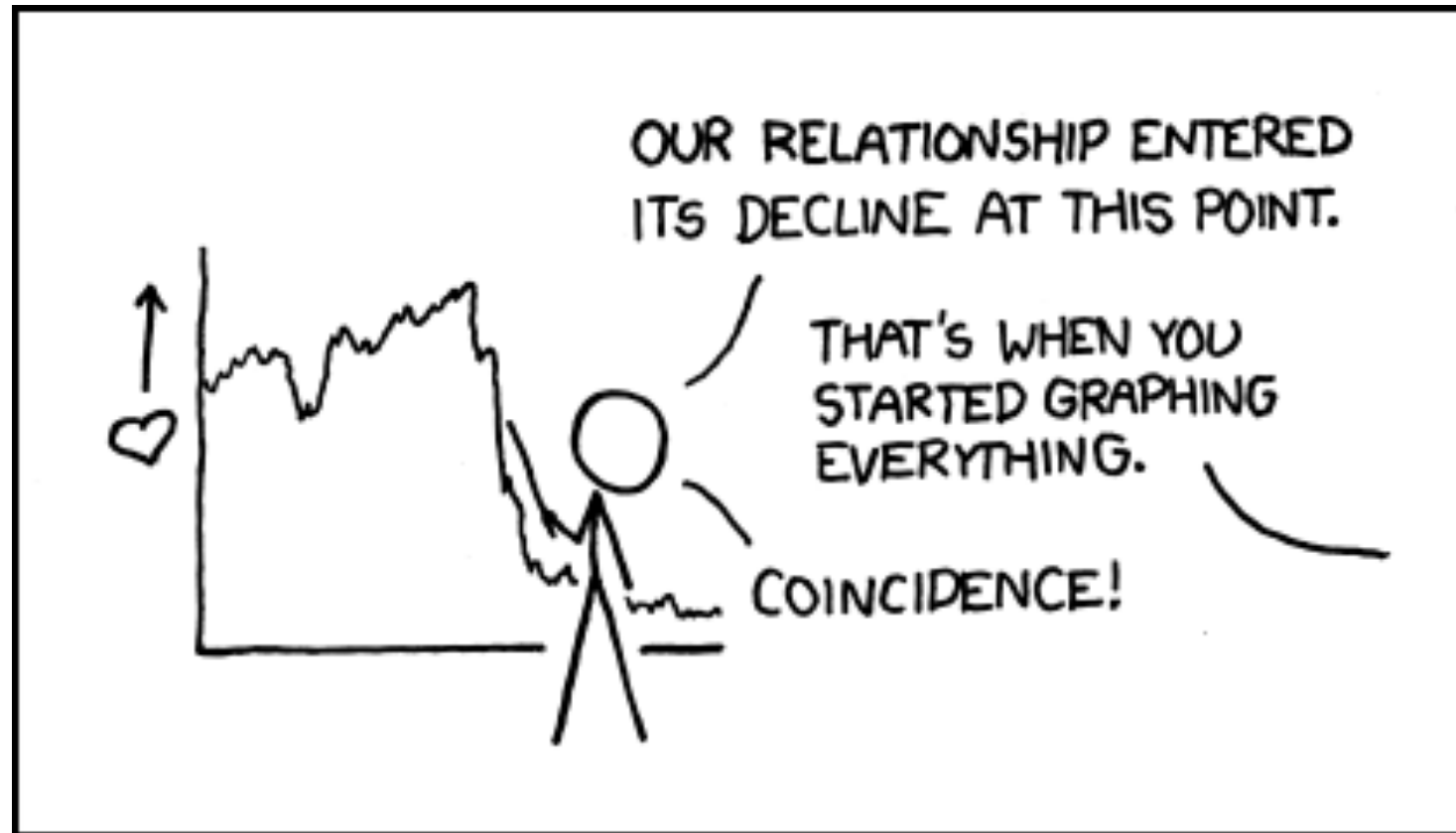
Thinking probabilistically

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Data comes from multiple systems

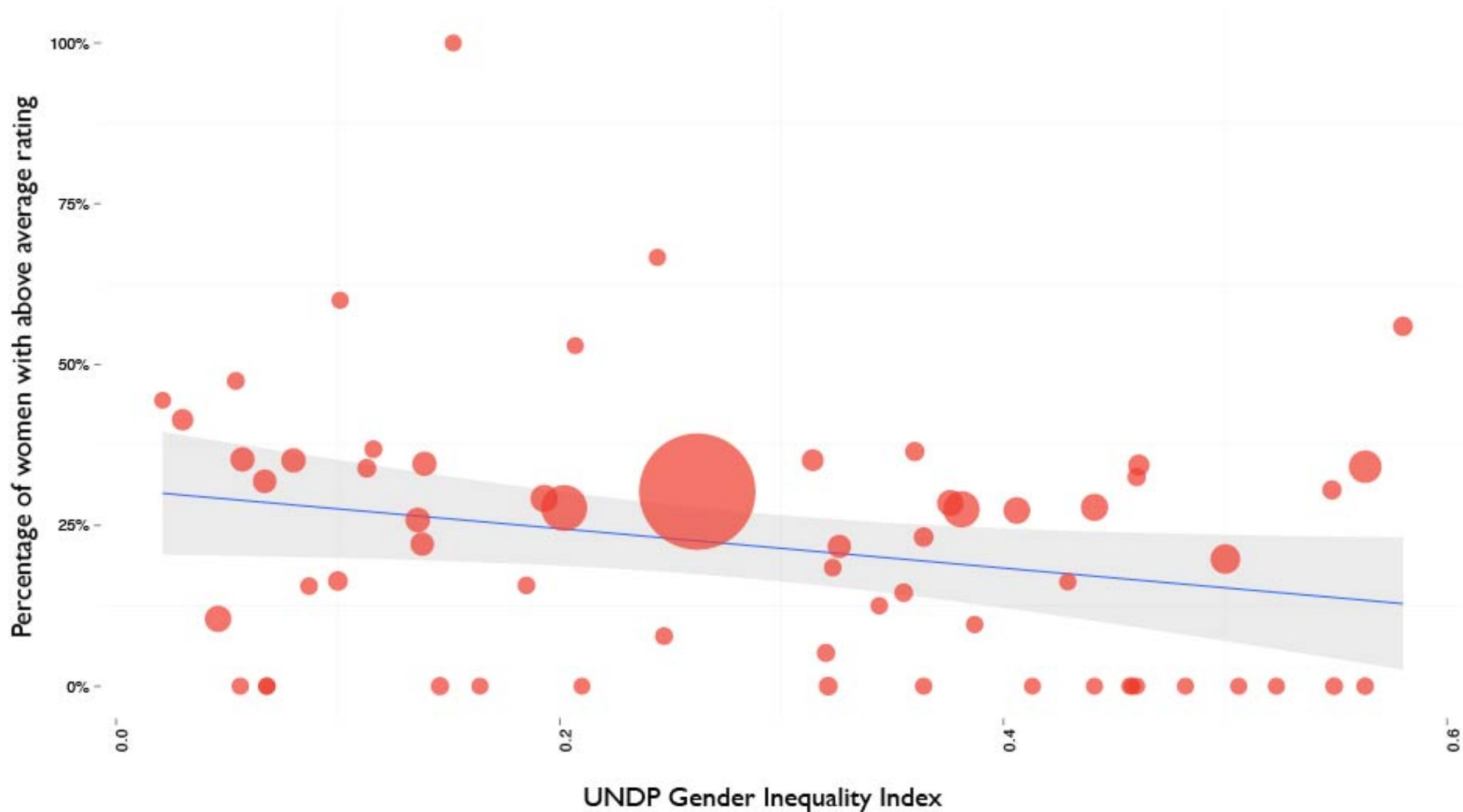


Data visualisation



Are performance ratings influenced by a country's attitude to gender equality?

Female ratings in a typical global employer. Bubbles show countries, scaled to reflect employee population



Visual Information-Seeking Mantra:
overview first, zoom and filter, then details
on demand.

Ben Schneiderman

Data visualisation to help people understand and explore

Adjust the weightings to personalise the ranking

1

By default the overall ranking gives no special weighting. To personalise the ranking based on what's important to you adjust the sliders below.

Customer weighting

Investor weighting

Jobseeker weighting

Media weighting

Message weighting

Society weighting

2

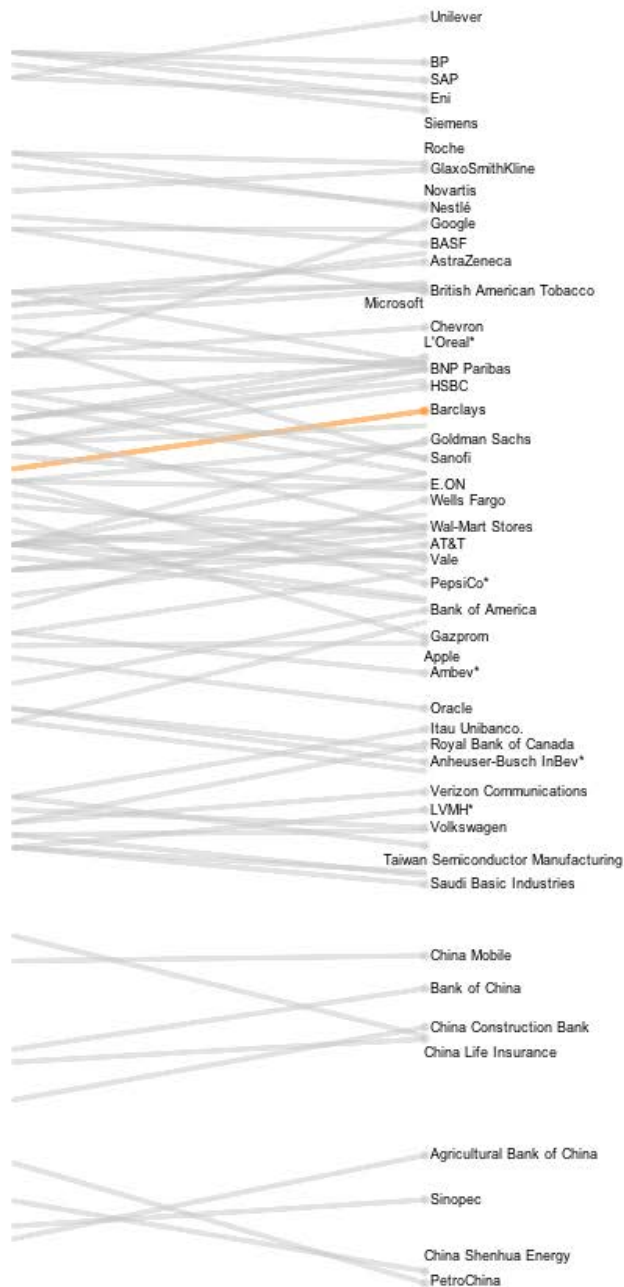
To highlight a company of interest please select it here:

Company to highlight

Barclays

Original position

Adjusted position



Winners and Losers

The following companies have changed the most based on your weightings.

Top 10 climbers

Google
Agricultural Bank of China
Wells Fargo
JPMorgan Chase & Co
Goldman Sachs
China Construction Bank
Royal Bank of Canada
Bank of China
Bank of America
Itau Unibanco.

Top 10 fallers

PetroChina
Industrial & Commercial Bank of China
Gazprom
Sanofi
China Shenhua Energy
PepsiCo*
Wal-Mart Stores
Vale
Daimler
Cisco Systems

Notes:

* Serving customers is proportional for all other metrics - ie Total other metrics / 248 x 32 = Customer

Where several companies have equal change they are sorted alphabetically.

Ranking trends 2008 to 2013

1

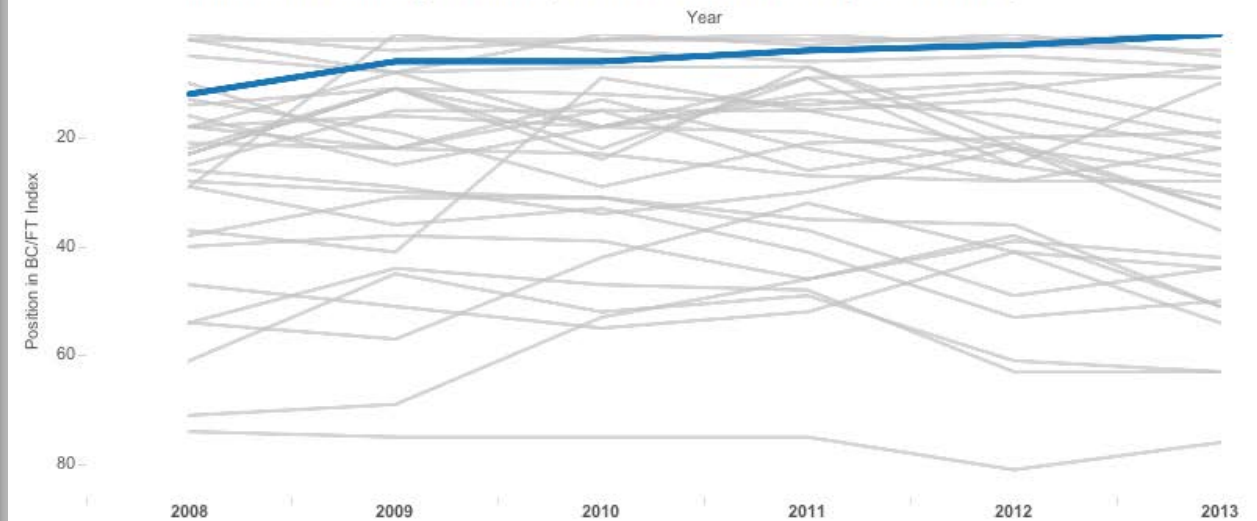
Choose a company

Eni

Thirty companies have been in the BC/FT Index for the last 6 years. Explore how they've changed over time.

Eni's performance over time

The blue line shows the ranking position of Eni against the other 29 firms who have been present in each ranking.

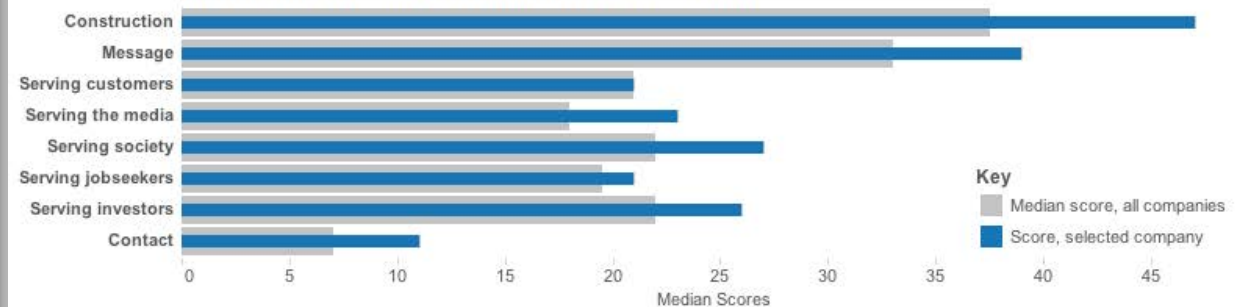


2

Select a year to show details of performance in each category

How Eni performed in 2013

Eni, in blue, shown against the median score of other firms in the 2013 ranking



Note

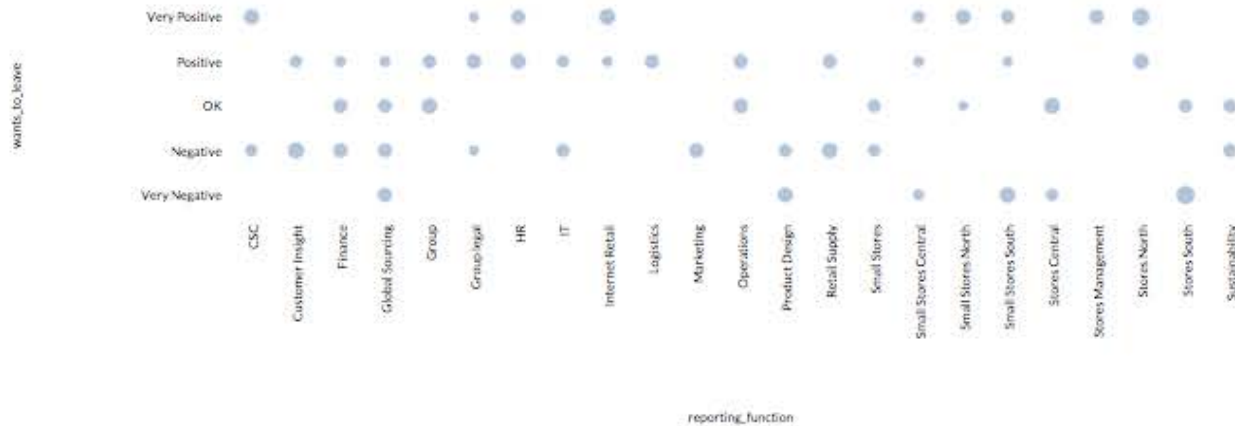
The Bowen Craggs methodology has evolved over time to account for changes in the way businesses communicate over the internet. To account for this comparisons are made to other participants in each year rather than absolute scores.

Text analytics & feedback

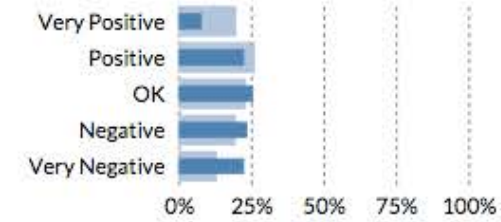
Comment Explorer

Selection: concept = Target

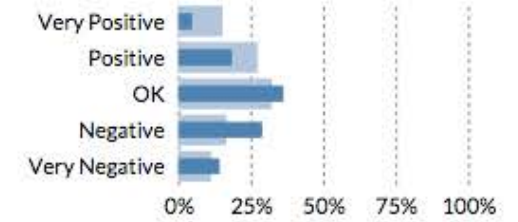
wants_to_leave reporting_function



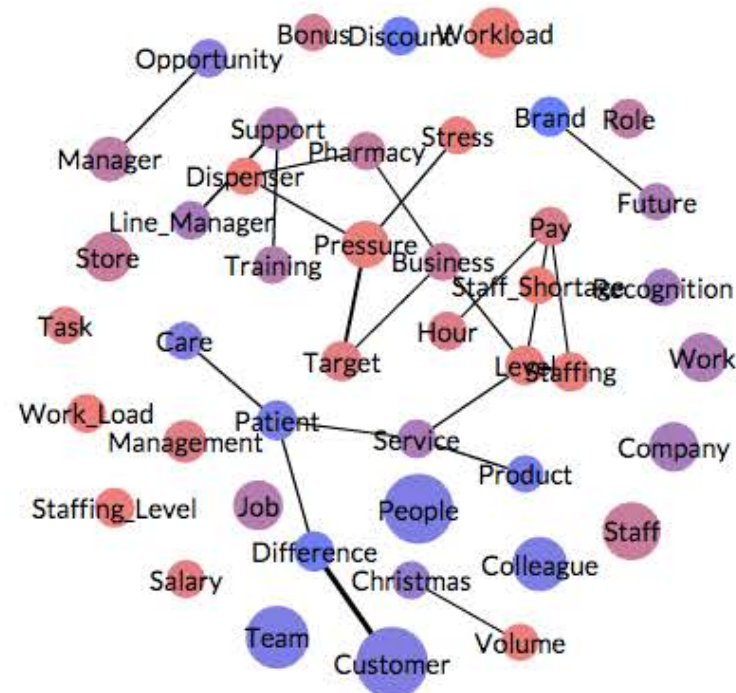
wants_to_leave



general_feeling



Comment Concept Graph



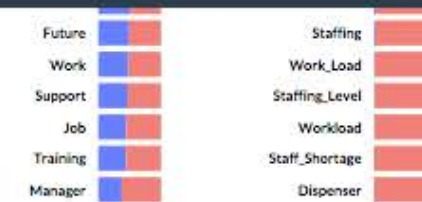
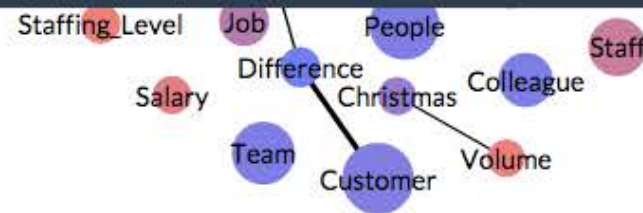
Concept Sentiment



Text analytics & feedback

Comment Explorer

Selection: concept = Target



Contexts

overload: **Workload** **Target** **Training** **Business** **Pharmacy** **Service** **Work_Load** **Company** **Pressure** **Customer**

problems/issues: **Staffing** **Work_Load** **Level** **Task** **Target** **Store** **Service** **Patient** **Pressure** **Management** **Support** **Staff** **Manager** **People**
Work **Company** **Job** **Customer** **Team**

proud of/enjoying: **Brand** **Job** **Role** **Work** **Company** **Customer** **Opportunity** **Product** **Recognition** **People** **Care** **Line_Manager**
Discount **Patient** **Manager** **Support** **Team** **Colleague** **Store**

shortage/lack: **Dispenser** **Staff** **Workload** **Work_Load** **Pressure** **Training** **Stress** **Bonus** **Hour** **Christmas** **Work**
Salary **Support** **Management** **Manager** **Staffing** **Recognition** **Target** **Business** **Care** **Pharmacy** **Line_Manager** **Opportunity**
Store **Level** **Staffing_Level** **Company** **Job** **Team** **People** **Customer**

stressful/frustrating: **Stress** **Work_Load** **Pharmacy** **Workload** **Staffing** **Staffing_Level** **Pressure** **Staff_Shortage** **Store** **Staff** **Target**
Work **Line_Manager** **Management** **Hour** **Pay** **Level** **Job** **Manager** **Colleague** **Team** **People** **Customer**

one_thing

Stores South

above
expectation



Managers are not concerned with looking after staff; only about hitting targets. Not through a lack of skill on their part; but because of the pressure on them by the company; who don't seem to get that if they

whats_good

Stores North

strongly
above
expectation



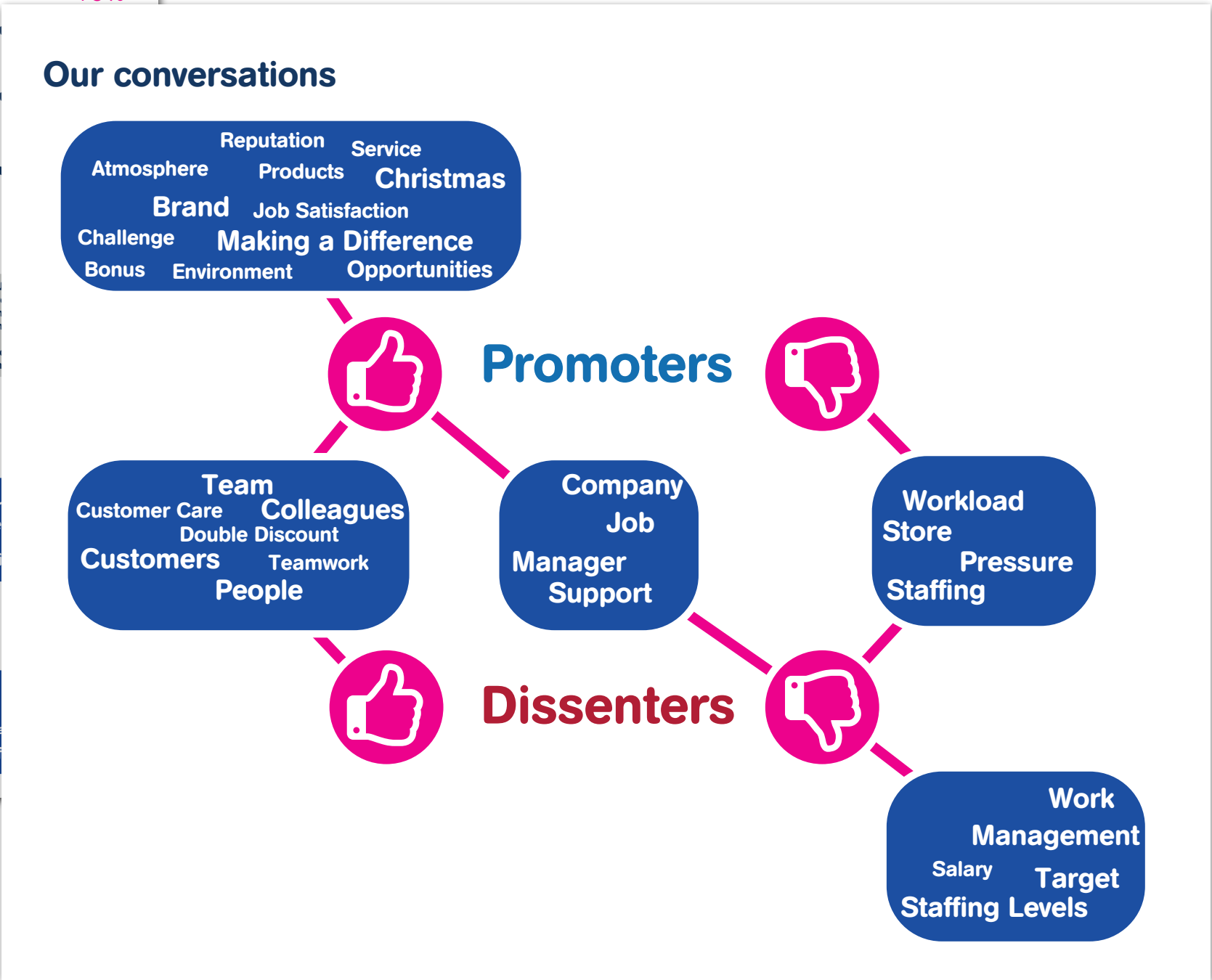
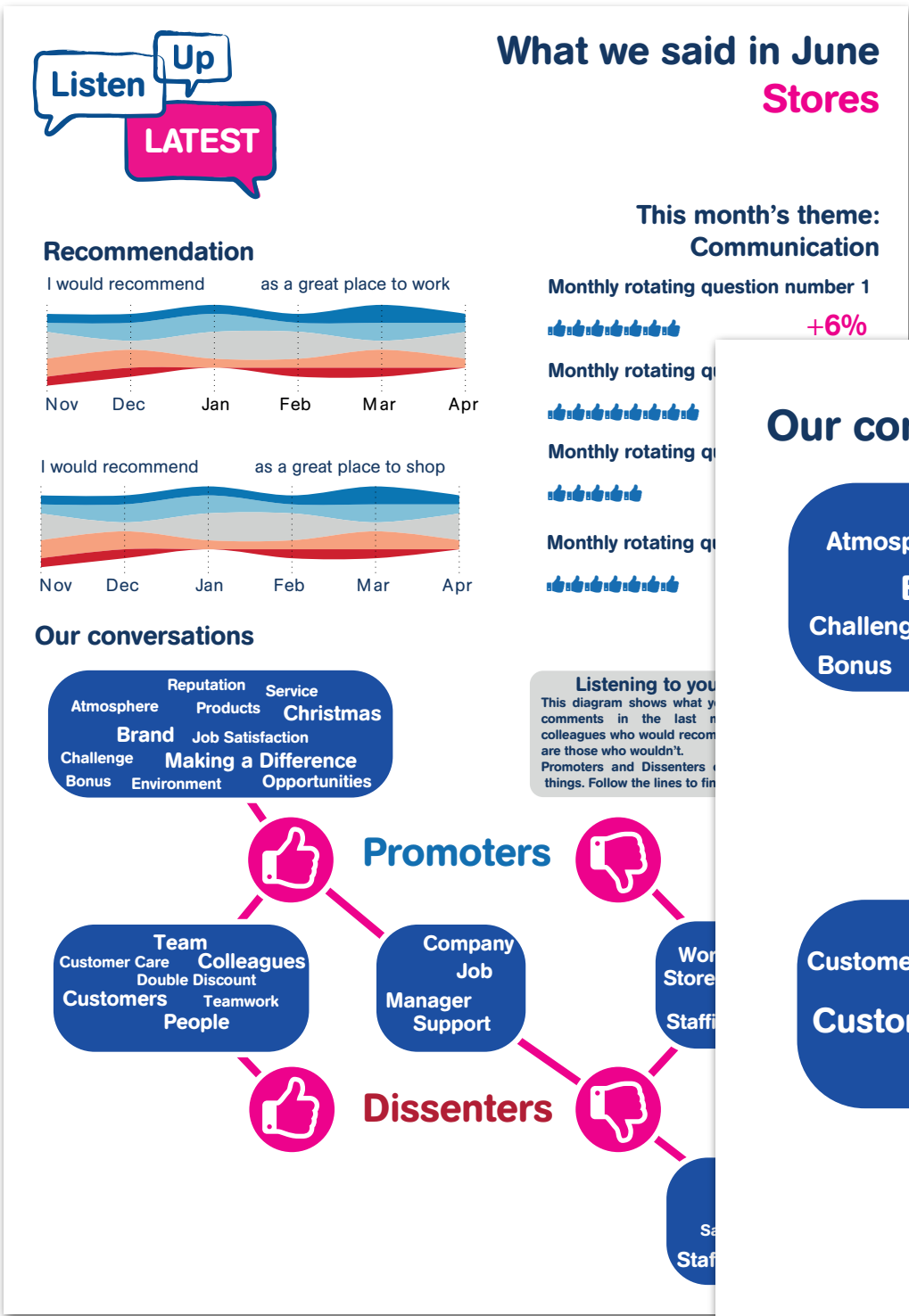
Internet Retail



Stores South



Text analytics & feedback



HOW TO DEVELOP ANALYST SKILLS

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Johns Hopkins University

Data Science

A Sequence of Courses: Learn to be a Data Scientist and Apply Your Skills in a Capstone Project

Final Capstone Project created with:



Overview

Certificate

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Start Specialization

In this course you will learn:

- Formulate context-relevant questions and hypotheses to drive data scientific research
- Identify, obtain, and transform a data set to make it suitable for the production of statistical evidence communicated in written form
- Build models based on new data types, experimental design, and statistical inference

In this Capstone Project, you will:

- Build a predictive data model for analyzing large textual data sets
- Clean real-world data and perform complex regressions
- Create visualizations to communicate your data analyses
- Build a final data product in collaboration with SwiftKey, award-winning developer of leading keyboard apps for smartphones

Develop your skills with competitions

CrowdANALYTIX

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Predict Cost of Worker's Compensation Claims

Predict cost of worker's compensation claims and identify relevant drivers

MODELING

START
12 MAY 15
2:00 PM

END
1 JUN 15
2:00 PM

RESULTS
25 JUN 15
2:00 PM

🏆 \$ 10000 👤 102 Solvers

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Description

Introduction

Workers' compensation is a form of insurance providing wage replacement and medical benefits to employees injured whilst in employment. Workers' compensation statutes are designed to ensure that any employee injured (ranging from short term harm to long term/permanent disablement) whilst on the job is not required to cover medical bills relating to their injury. Additionally, the employee is provided with monetary awards to cover the loss of wages as a direct consequence of the accident and to compensate in the event of any permanent physical impairment. The law aims to eliminate the need for litigation between employee and the employer in the event of an injury.

In most US states it is a legal requirement for employers to have a workers' compensation insurance policy to protect themselves against potential compensation claims for workplace injuries. As such, insurance companies provision substantial sums of money to pay out on future workers' compensation claims.

Objective

To analyse and quantify the relationship and/or correlation between individual claim characteristics to determine which claim characteristic(s) can be used to predict the ultimate cost of a workers' compensation claim.

Data

The underlying data represents claim (or loss) information for employees injured (i.e. claimants) while at work during

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Anoop Venkatachala

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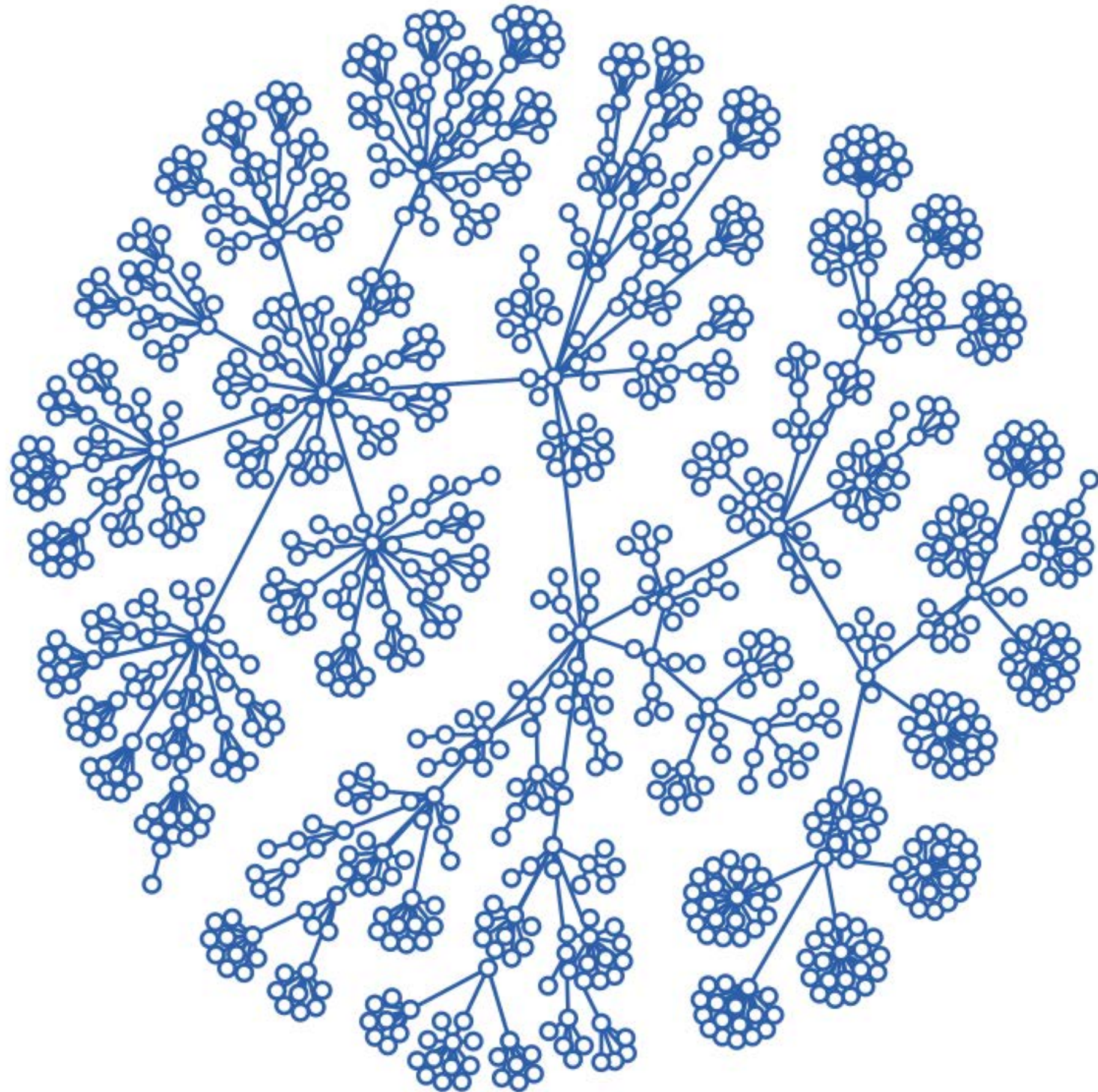


Subhamoy Ganguly

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